

Immunomodulators

Substances that affect the immune system by stimulating it or suppressing it.

- Immunosuppressants - Decrease or block the body's immune response.
AI disorders, organ transplant rejection
- Immunostimulants - Improve immune response
Immunodeficiency disorders (HIV/AIDS), some cancer treatment

Corticosteroids - dexamethasone, prednisone	
Low dose - inhibit inflammatory WBCs (anti-inflammatory) High dose - block T&B cell production (immunosuppression)	
Side Effects & Adverse Effects • Hyperglycemia • Infection risk • GI • Insomnia • Cushingoid appearance • Swelling	Education • Take with food • Taper off • No live vaccines

Calcineurin Inhibitors - cyclosporine	
Organ transplant anti-rejection (taken long term, often life) Some AI disorders	
Side Effects & Adverse Effects • Nephrotoxicity - kidney • Infection risk • Myelosuppression - anemia, infection, bleeding • Gum hyperplasia • GI • Flu-like	Education • Kidneys • Infection • Electric razor use • Fall risk • Soft toothbrush • No blood thinners

Biologic Agents Monoclonal antibodies - trastuzumab	
Genetically engineered - highly targeted to proteins Cancer, AI, RA, Chrons	
Side Effects & Adverse Effects • Fatigue • Flu-like • Less S/E than chemo	Education • Infection • Site infection - IV admin

Biologic Agents Disease Modifying Antirheumatic Drugs (DMARDs) - Methotrexate	
Decreases inflammation, prevents joint damage Inhibits the enzyme for metabolism and increased adenosine molecules RA gold standard drug	
Side Effects & Adverse Effects • Liver toxicity • Infection	Education • Limit alcohol • No pregnancy • 1x weekly dose - be careful • 3-12 weeks to work • No live vaccines

Biologic Agents Disease Modifying Antirheumatic Drugs (DMARDs) - Hydroxychloroquine	
Decreases inflammation, prevents joint damage Interferes with immune system communication RA, lupus, malaria	
Side Effects & Adverse Effects • Retinal damage • Rash • Diarrhea	Education • Eye exams every 6-12 months • 3-12 weeks to work • Report vision changes immediately

Biologic Agents Biologic DMARDs/TNF Blocker - Adalimumab	
Interrupt signals that activate inflammation Can be used with traditional DMARDS (Metho) or when traditional not working Inflammatory AI arthritis	
Side Effects & Adverse Effects • Infection • Site Infection • Malignancy (lymphoma) • TB	Education • Infection • SubQ - infection • No live vaccines • TB test - at start, every year • Blood counts

Medications to Treat Inflammation

Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) - 1st Generation Ibuprofen, Aspirin	
Decrease inflammation, pain and fever Inhibits COX-1 and COX-2; prevents synthesis of prostaglandin	
Side Effects & Adverse Effects • GI bleeding • Kidneys (ibuprofen) • Bleeding (aspirin) • Reye's Syndrome (aspirin)	Education • Bleeding • Dosage

Non-Steroidal Anti-Inflammatory Drugs (NSAIDs) - 2nd Generation Celecoxib	
Decrease inflammation, pain and fever COX-2 inhibited only	
Side Effects & Adverse Effects • CV events • MI • Stroke	Education • CV events

Acetaminophen	
Pain and fever reduction Block production of prostaglandins	
Side Effects & Adverse Effects • Hepatotoxicity - liver Antidote: acetylcysteine	Education • Alcohol use • Look for this in other OTC meds • 4g daily does

Glucocorticoids - Prednisone	
Binds to glucocorticoid receptors, suppressing inflammatory cytokines Inflammatory and AI diseases	
Side Effects & Adverse Effects • Hyperglycemia • Infection risk • GI • Insomnia • Cushingoid appearance	Education • Take with food • Taper off • No live vaccines

• Swelling	
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Antineoplastics

Treatments for Cancer

- Chemotherapy - Targets rapidly dividing cells, including healthy cells.
- Hormone Therapy
- Biological Therapy

Myelosuppression

- Bone Marrow Suppression
- Neutropenia - Low WBC; Infection
- Thrombocytopenia - Low Platelets; Bleeding
- Anemia - Low RBC; Fatigue

Alkylating Agents Nitrogen Mustards - Cyclophosphamide	
Chemo Leukemia, lymphoma	
Side Effects & Adverse Effects • Hemorrhagic cystitis - lining of the bladder infection (blood in urine) • GI	Education • Hydration - flush toxins and decrease risk

Anthracyclines - Doxorubicin Hydrochloride	
Chemo Red Devil	
Side Effects & Adverse Effects • Red/orange urine • Cardiotoxicity	Education • Red/orange bodily fluid for days

Vinca Alkaloids - Vincristine	
Chemo Mitotic inhibitors	
Side Effects & Adverse Effects • Neurotoxicity - peripheral neuropathy (tingling/numbness of hands and feet)	Education • Peripheral neuropathy • Infection

Folic Acid Analogs - Methotrexate	
Chemo - mitotic inhibitor	
Side Effects & Adverse Effects • Liver toxicity • Infection	Education • Limit alcohol • No pregnancy • 1x weekly dose - be careful • 3-12 weeks to work • No live vaccines

Estrogen Receptor Blockers - Tamoxifen	
Hormonal Therapy Breast cancer & estrogen dependent cancers	
Side Effects & Adverse Effects • Anti-estrogen - hot flash, etc. • Uterine cancer • Blood clots (thromboembolism - moving clot)	Education • Menopausal effect • Clotting • Risks

Monoclonal Antibodies - Trastuzumab	
Biological Therapy HER2 positive breast cancer	
Side Effects & Adverse Effects • Cardiotoxicity - heart failure signs	Education • Endocardiogram before and periodically

Anti-Infectives

- Bacterial: UTI, cellulitis = antibiotics
- Fungal: yeast infection, oral thrush = antifungals (can be hard to treat with adverse effects)
- Viral: influenza, HIV = antivirals (inhibit replication usually)
- Parasitic: malaria = antiparasitic

Antibacterials

Tetracyclines - Tetracycline	
Respiratory tract infection; some STIs Inhibits bacterial protein synthesis - broad spectrum coverage	
Side Effects & Adverse Effects • GI • Photosensitivity • Permanent tooth discoloration under 8yrs (do not take)	Education • Pregnancy (avoid especially in 2nd and 3rd trimesters) • Sunscreen and limit sun exposure • Dairy and iron reduce absorption

Lincosamides - Clindamycin	
Skin & Soft Tissue Infection Inhibits bacterial protein synthesis (anaerobic bacteria and G+)	
Side Effects & Adverse Effects • GI - N, V, D • Clostridioides difficile diarrhea risk	Education • Take with a full glass of water • Complete full course • Pregnancy 1st trimester

Sulfonamide Derivatives - Trimethoprim-Sulfamethoxazole (TMP-SMX)	
UTI 1st line agent Disrupts folic acid pathway (2 drugs, synergism)	
Side Effects & Adverse Effects • GI - N, V, D • Stevens-Johnson Syndrome - Rare, life threatening skin and mucous	Education • Report new rash immediately

membranes rash	
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Aminoglycosides - Gentamicin	
Sepsis, UTI Inhibit bacterial protein synthesis	
Side Effects & Adverse Effects • Nephrotoxicity • Ototoxicity (ears)	Education • Report ear ringing, hearing changes, dizziness, vertigo • Kidney impairment • Pregnancy

Beta-Lactams - Amoxicillin, Cephalosporins	
Amoxicillin/Penicillin: Strep, pneumonia, skin/soft tissue infection Cephalosporins: UTIs, meningitis Inhibits bacterial wall synthesis	
Side Effects & Adverse Effects • GI • Allergy - penicillin	Education • Allergy and other allergies

Fluoroquinolones - Ciprofloxacin	
UTIs (when other options are not available) Target bacterial DNA replication	
Side Effects & Adverse Effects • GI • Risk of tendinitis and tendon rupture	Education • No previous tendon disorders • No antacids or dairy within two hours of dosage - absorption interference

Glycopeptides - Vancomycin	
MRSA, Complicated infection	
Side Effects & Adverse Effects • Infusion reaction - turn red • Nephrotoxicity - liver impairment	Education • Intense med

• Ototoxicity	
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Rifamycins - Rifampin	
TB	
Side Effects & Adverse Effects • Hepatotoxicity • Orange/red body fluids	Education • fluids

Antifungals

Azole Derivatives - Fluconazole	
Vaginal Yeast infection, oral thrush	
Side Effects & Adverse Effects • Hepatotoxicity • QT interval prolonged • GI	Education • Antacids 1 hr before or 2hr after • No alcohol

Polyene Macrolides - Amphotericin B	
Serious, life threatening fungal infection High alert - amphoterrible	
Side Effects & Adverse Effects • Nephrotoxicity • Electrolyte imbalance • Cardiac arrest	Education • Severe effects

Antiparasitic

Antiprotozoal Agents - Metronidazole	
Anaerobic bacterial infection, certain parasitic infections, C. Diff diarrhea	
Side Effects & Adverse Effects • Peripheral neuropathy (tingling/numb) finger and toes • GI - N • Metallic taste in mouth	Education • No alcohol within last 72 hrs • Large pills

Antivirals

Neuraminidase Inhibitors - Oseltamivir Phosphate	
Influenza A & B (tamiflu) Most effective in early illness (within 48hrs)	
Side Effects & Adverse Effects •	Education •

Nucleoside Analogues - Acyclovir Sodium	
Herpes infection (treat/suppress outbreaks)	
Side Effects & Adverse Effects • Nephrotoxicity (especially via IV) • GI - N	Education •

Nucleoside Reverse Transcriptase Inhibitors - Zidovudine	
HIV (part of regimen)	
Side Effects & Adverse Effects •	Education • Adherence • Take for life

Side Effects & Adverse Effects	Education

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